L15 HW

- Due Mar 5 at 11:59pm
- Points 20
- Questions 20
- Time Limit None
- Allowed Attempts 3

Instructions

You get three attempts on all homework quizzes.

In all quizzes and homeworks in this course, round your answers to THREE DECIMAL places unless otherwise indicated.

Attempt History

	Attempt	Time	Score
KEPT	Attempt 3	3 minutes	20 out of 20
LATEST	Attempt 3	3 minutes	20 out of 20
	Attempt 2	5 minutes	19 out of 20
	Attempt 1	38 minutes	16 out of 20

Score for this attempt: 20 out of 20
Submitted Mar 4 at 12:57pm
This attempt took 3 minutes.
Correct answer

Question 1
1 / 1 pts
Which of the following is NOT one of the 3 basic principles of experimental design?

- (selected) Response units
- Randomization
- Replication

- Local control

An experiment to assess the effect of three types of coating on the life span of a certain type of wood. Use this information for all the parts.

Correct answer

Question 2

1 / 1 pts

Part 1: How many treatments are involved in this study?

3

3 (with margin: 0)

Correct answer

Question 3

1 / 1 pts

Part 2: The experiment will be carried out in 3 different locations. It is known that specific weather (e.g. temperature and humidity) also affect the life span of the wood. Each location is randomly assigned a coating. True or False: The location is confounded with the type of coating.

- (selected) True
- False

Correct answer

Question 4

1 / 1 pts

Part 3: What key principle of experimental design is missing from the description of this wood coating experiment as described above? ***(check all that apply)***

- [x] replication
- [ ] randomization
- [x] Local control
- [ ] None of the above

Correct answer

Question 5

1 / 1 pts

What type of variability do we typically hope to see in the response variable of an experiment?

- (selected) Planned, systematic variability that comes from differences in the conditions of interest
- Unplanned, systematic variability arising from variability in the experimental units or experimental process

- None: we don't want any variability in response variable.

Correct answer

Question 6

1 / 1 pts

One of the 3 principles of a good experimental design is replication. What does replication refer to in this context?

- Gathering more than one observation per treatment
- Having an independent group of researchers repeat the study and get the same result
- Assigning an experimental unit to receive more than one treatment
- Taking more than one measurement on each experimental unit

Correct answer

Question 7

1 / 1 pts

Which of the following is a reason for randomizing?

- randomizing converts unplanned, systematic variability into chance-like variability
- Randomizing controls the effects of lurking variables
- Randomizing reduces the opportunities for bias to enter the experiment
- All of the above
- None of the above

An experiment was conducted to study the effect of temperature and type of oven on the life of the component being tested. The numbers recorded in the table below represent the number of days the component lasted before it broke. Use this table to answer all the parts that follow.

Temperature (degrees)	Oven			
	O_1	O_2	O_3	O_4
500	227	214	225	260
	221	259	236	229
550	187	181	232	246
	208	179	198	273
600	174	198	178	206
	202	194	213	219

Correct answer

Question 8

1 / 1 pts

Part 1: Which of the following is the factor? **(Mark all that apply)**

- [ ] Components
- [ ] Number of days
- [x] Oven
- [x] Temperature

Correct answer

Question 9

1 / 1 pts

Part 3: Which of the following is the response variable?

- ( ) Components
- (•) Number of days
- ( ) Oven
- ( ) Temperature

Correct answer

Question 10

1 / 1 pts

Part 3: Which of the following are the experimental units?

- (•) Components
- ( ) Number of days
- ( ) Ovens
- ( ) Temperatures

Correct answer

Question 11

1 / 1 pts

Part 4: Is this a balanced design?

- (•) Yes
- ( ) No
- ( ) There is no way to tell from the information given.

Correct answer

Question 12

1 / 1 pts

Part 5: How many replications are there in each treatment?

2

2 (with margin: 0)

Correct answer

Question 13

1 / 1 pts

Part 6: How many treatments are there?

12

12 (with margin: 0)

Correct answer

Question 14

1 / 1 pts

Part 7: What is the first treatment mean?

- 203.167
- 227
- 224
- 233.875

Three strains of rats were studied under 2 environmental conditions for their performance in a maze test. For each treatment, the error scores (performance) for eight different rats were recorded as in the data below. Use this information to answer all parts.

	Strain					
Environment	Bright		Mixed		Dull	
Free	28	12	33	83	101	94
	22	23	36	14	33	56
	25	10	41	76	122	83
	36	86	22	58	35	23
Restricted	72	32	60	89	136	120
	48	93	35	126	38	153
	25	31	83	110	64	128
	91	19	99	118	87	140

Correct answer

Question 15

1 / 1 pts

Part 1: How many rats were tested here?

48

48 (with margin: 0)

Correct answer

Question 16

1 / 1 pts

Part 2: How many treatment combinations are there?

6

6 (with margin: 0)

Correct answer

Question 17

1 / 1 pts

Part 3: How many replications were done in each treatment?

8

8 (with margin: 0)

Correct answer

Question 18

1 / 1 pts

Part 4: What is the response variable?

- Rats
- Strain
- Environment
- Performance
- None of the above

Correct answer

Question 19

1 / 1 pts

Part 5: Which of the following is NOT a factor in this experiment?

- Strain
- Environment
- Rats
- All the above are factors.

Correct answer

Question 20

1 / 1 pts

Part 6: Is this a fixed or a random factor model?

- Fixed
- Random

Quiz Score: 20 out of 20

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